STAT 570 Probabilistic Machine Learning
Assignment 3-2

For this assignment, you can use either R or Python. There is no automatic testing component. Submit your code and your write-up in PDF format. Submit all code as `problem1.py` (or `problem1.R`).

Problem 1: Particle filter (100 points)

We will reproduce the results from "P. Fearnhead, P. Clifford, On-line inference for hidden Markov models via particle filters. J. R. Stat. Soc. Series B Stat. Methodol. 65, 887–899 (2003)."

The well-log data is provided in `welldata.csv`. Your task is to reproduce Figure 2 in Fearnhead and Cliffod (2003).

- An implementation of the optimal sampling algorithm described in Section 4 of the paper is provided in `problem1.py`.
- Implement a particle filter for online inference in a hidden Markov model (HMM). Use the model specification described in Section 3 of the paper.
- Apply your particle filter to the `welldata.csv` data.
- Reproduce Figure 2, which shows the posterior mean of the hidden states at each time point, overlaid on the observed data.

For partial credit:

- Let $I_t = (S_t, O_t)$. For $t = 1$, derive the formula for α, γ for all possible vlues of S_t and O_t, i.e., $(1,1), (1,2), (2,1), (2,2)$. Provide the formula in your write-up for partial credit.
- Repeat the derivation for $t = 2$ and generalize for any $t > 1$. Provide the formula in your write-up for partial credit.
- Explain how to update the change point τ.